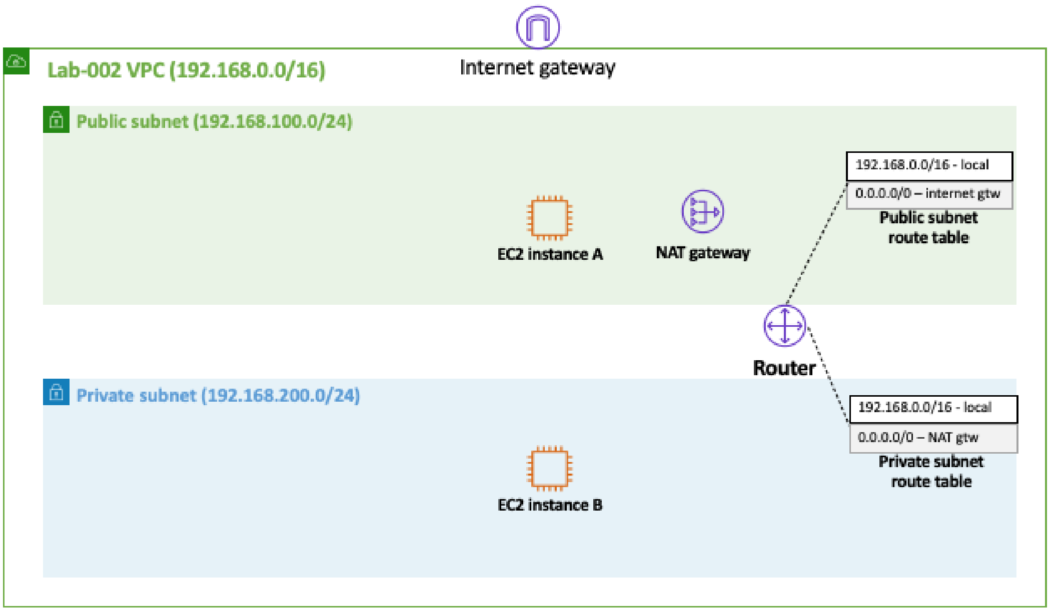
Lab-003

A Single EC2 Instance in a Private Subnet + Bastion Host + NAT Gateway

Goal

This lab illustrates how an EC2 instance running in a *private* subnet can be configured to access the internet with the help of a service called Nat gateway.

Architecture Diagram



Overview

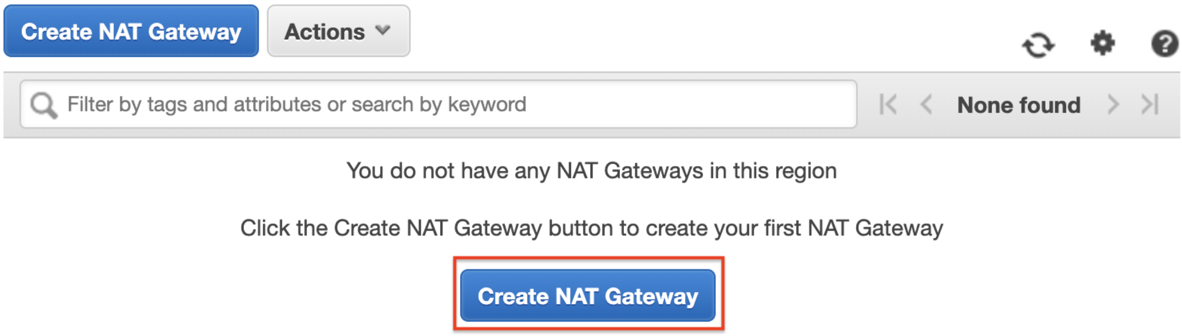
In order to achieve the goal of this lab, you will have to go through the following steps:

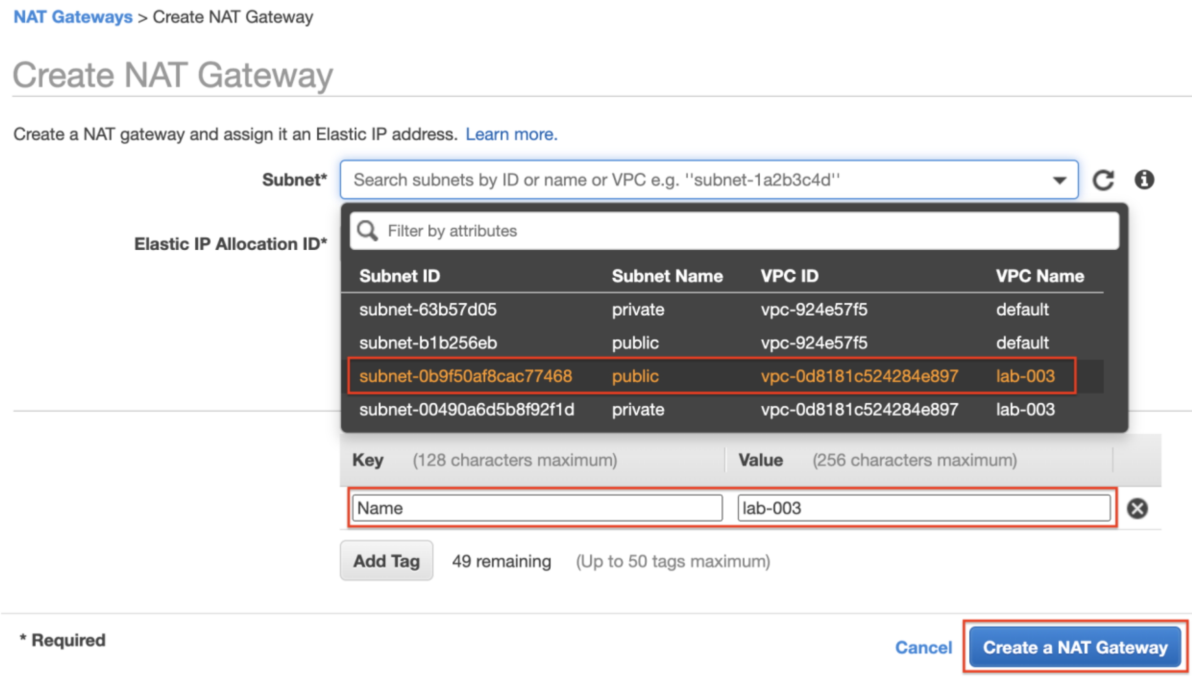
Step 1 - Run Lab-002

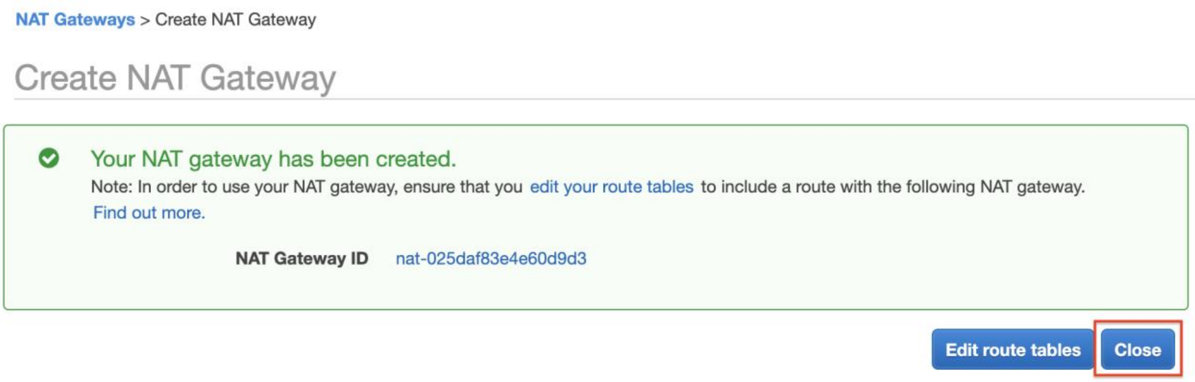
Repeat all of the steps described in lab-002, making sure that any reference to lab-002 are replaced by lab-003.

Step 2 - Create a NAT Gateway

Create a NAT gateway in the public subnet. Note that an elastic public IP must be allocated before a NAT gateway can be created

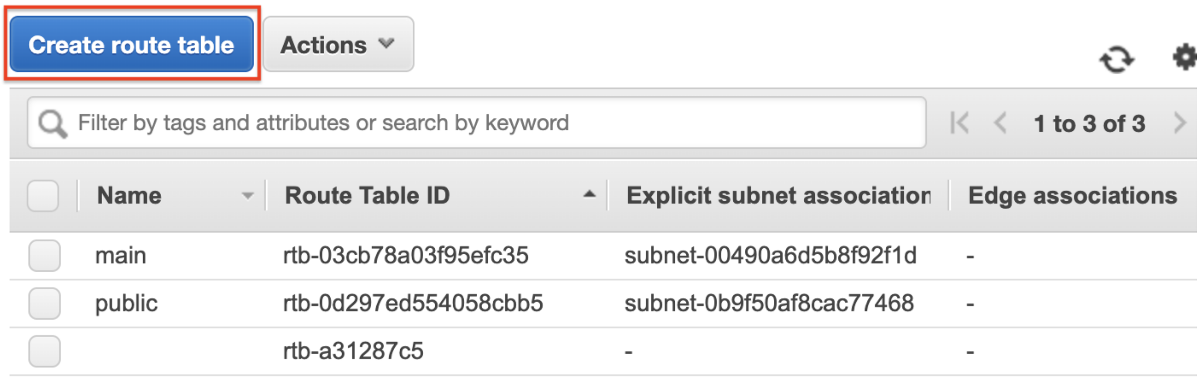


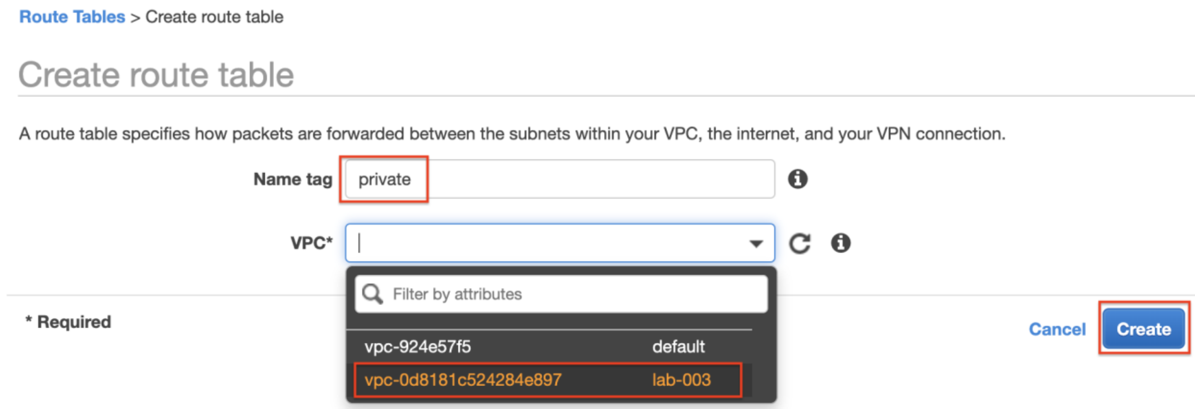


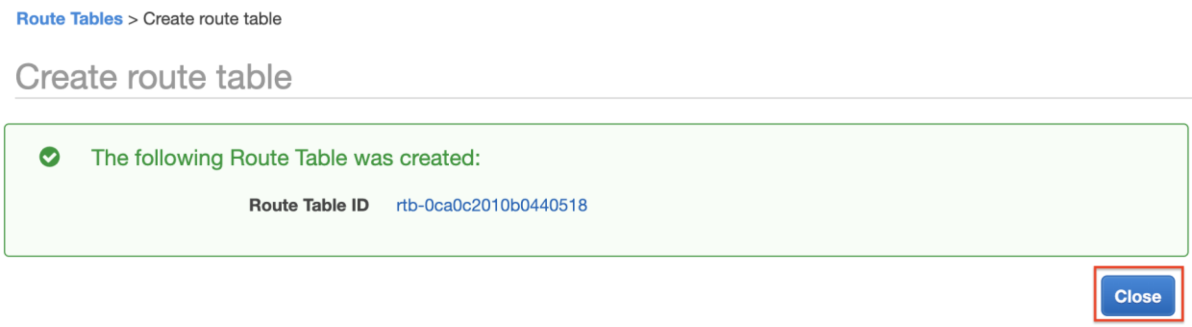


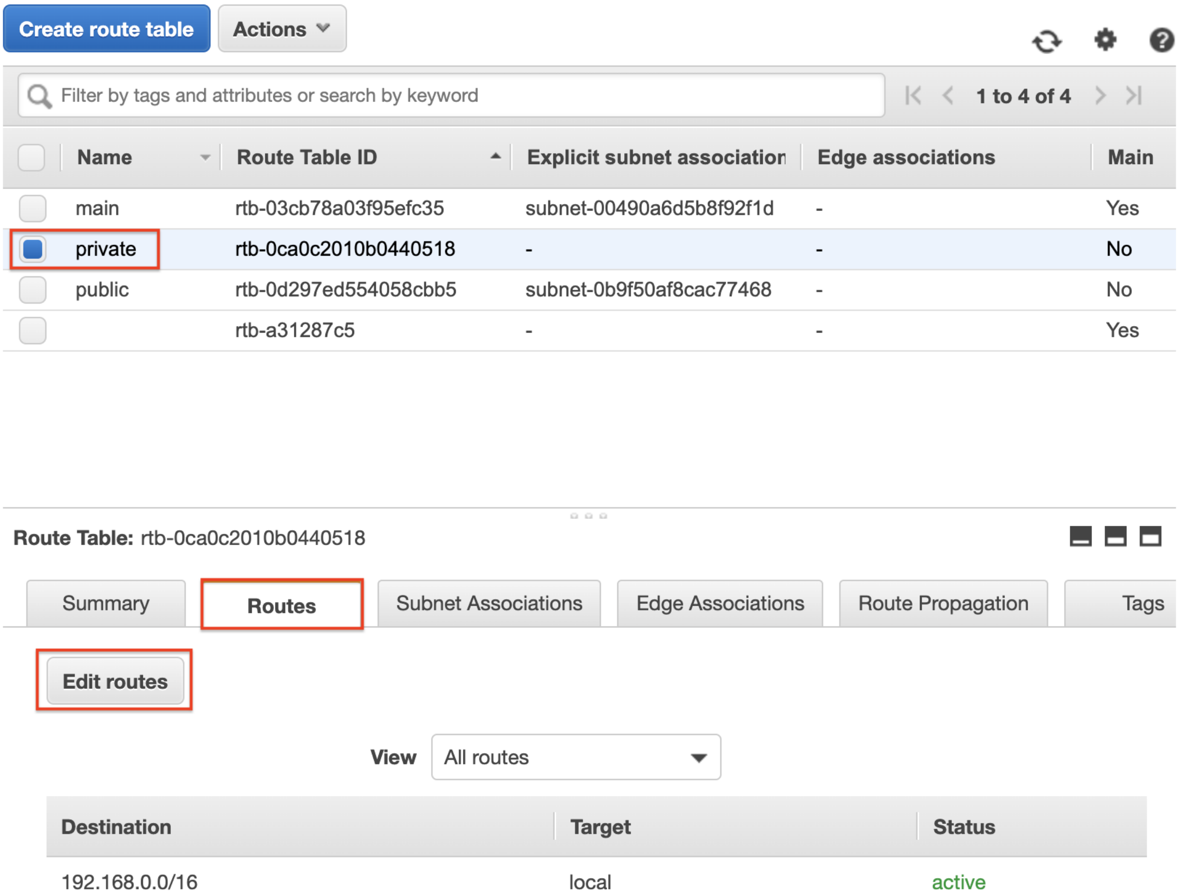
Step 3 - Create a Route Table

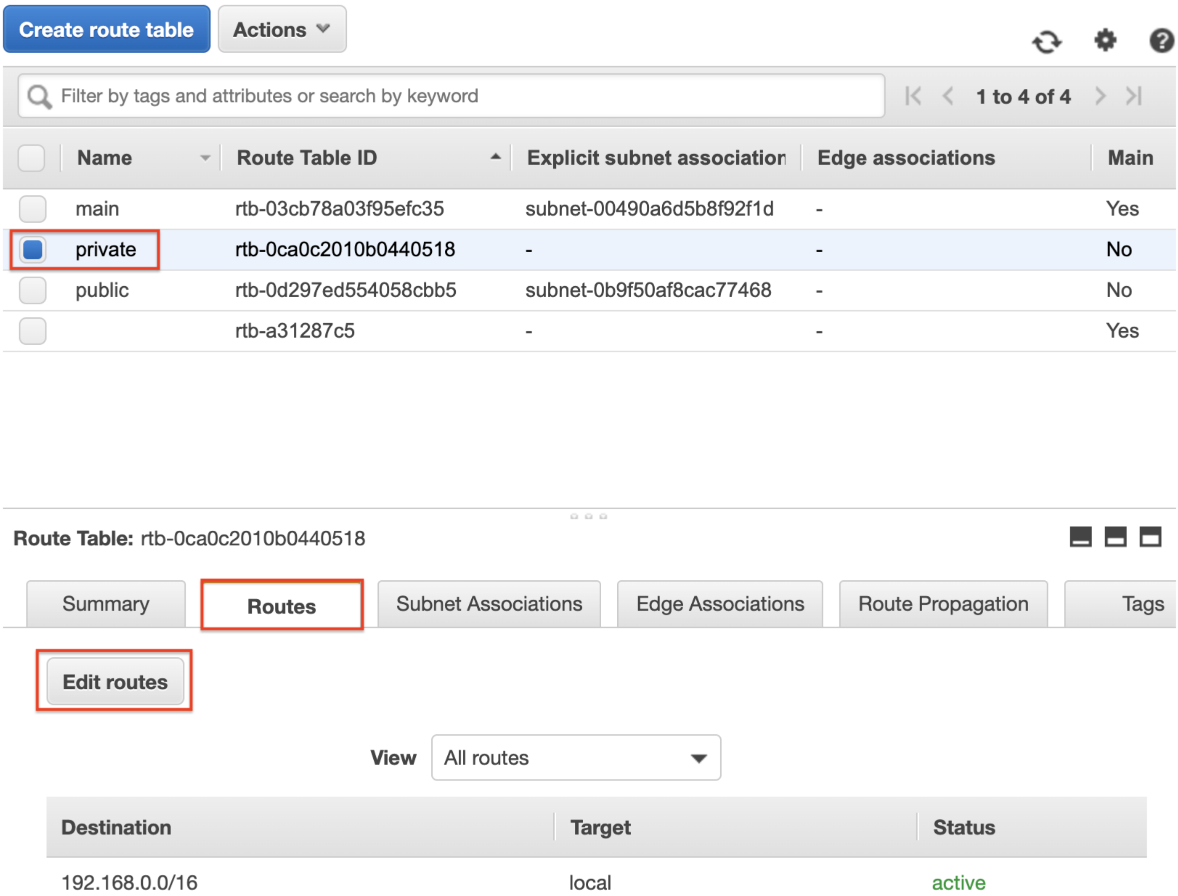
Create a new route table with the *Name tag* private and with a default route to the NAT gateway created in step 2.

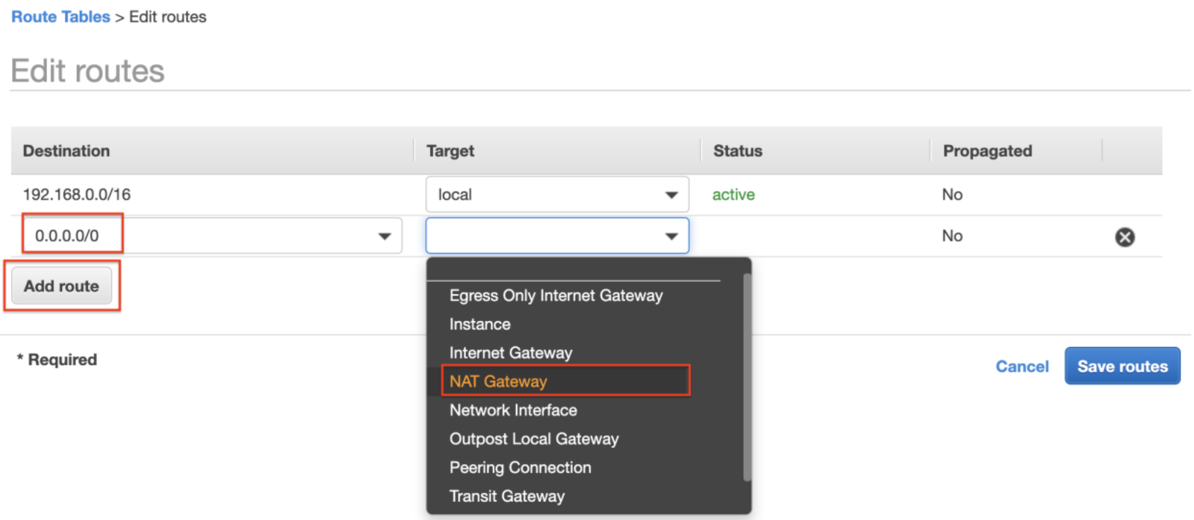

Create route table
Actions
Filter by tags and attributes or search by keyword
1 to 3 of 3
Name
Route Table ID
Explicit subnet association
Edge associations
main
rtb-03cb78a03f95efc35
subnet-00490a6d5b8f92f1d
-
public
rtb-0d297ed554058cbb5
subnet-0b9f50af8cac77468
-
rtb-a31287c5
-
-

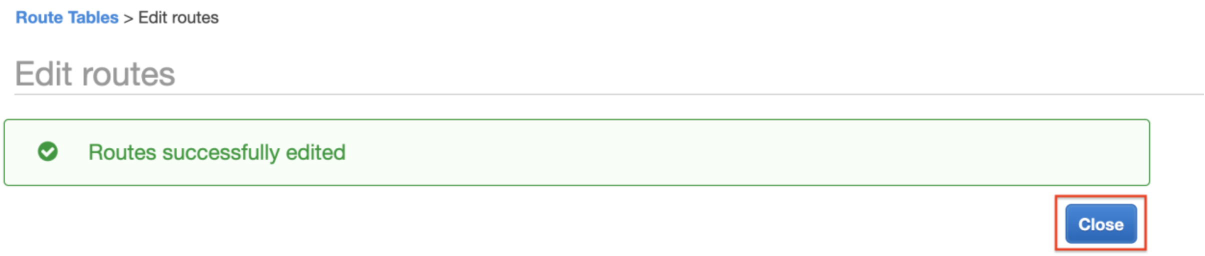

Route Tables > Create route table
Create route table
A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.
Name tag
private
VPC*
Filter by attributes
vpc-924e57f5
default
vpc-0d8181c524284e897
lab-003
* Required
Cancel
Create

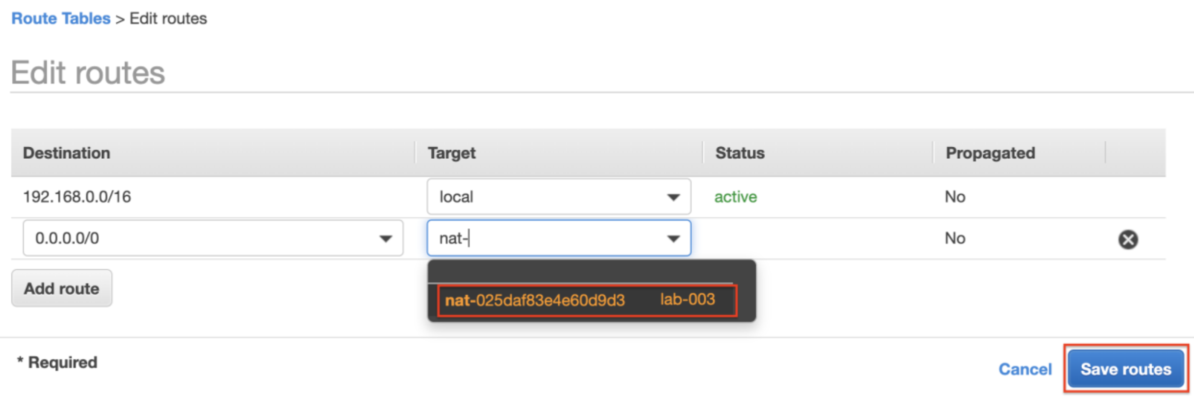

Route Tables > Create route table
Create route table
The following Route Table was created:
Route Table ID
rtb-0ca0c2010b0440518
Close

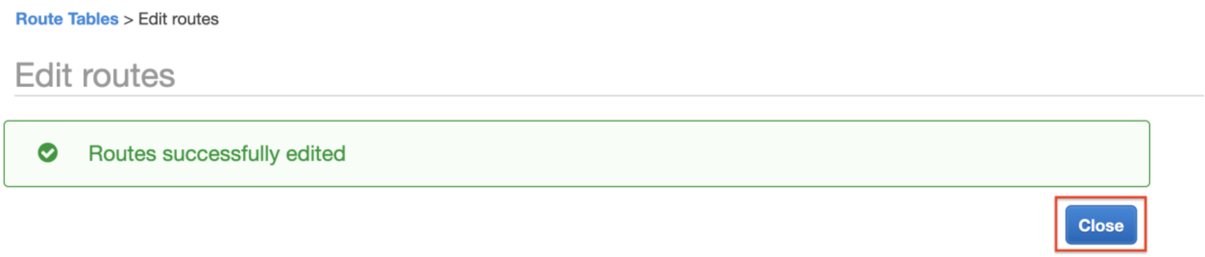

Create route table
Actions
Filter by tags and attributes or search by keyword
1 to 4 of 4
Name
Route Table ID
Explicit subnet association
Edge associations
Main
main
rtb-03cb78a03f95efc35
subnet-00490a6d5b8f92f1d
-
Yes
private
rtb-0ca0c2010b0440518
-
-
No
public
rtb-0d297ed554058cbb5
subnet-0b9f50af8cac77468
-
No
rtb-a31287c5
-
-
Yes
Route Table: rtb-0ca0c2010b0440518
Summary
Routes
Subnet Associations
Edge Associations
Route Propagation
Tags
Edit routes
View
All routes
Destination
Target
Status
192.168.0.0/16
local
active


Create route table
Actions
Filter by tags and attributes or search by keyword
1 to 4 of 4
Name
Route Table ID
Explicit subnet association
Edge associations
Main
main
rtb-03cb78a03f95efc35
subnet-00490a6d5b8f92f1d
-
Yes
private
rtb-0ca0c2010b0440518
-
-
No
public
rtb-0d297ed554058cbb5
subnet-0b9f50af8cac77468
-
No
rtb-a31287c5
-
-
Yes
Route Table: rtb-0ca0c2010b0440518
Summary
Routes
Subnet Associations
Edge Associations
Route Propagation
Tags
Edit routes
View
All routes
Destination
Target
Status
192.168.0.0/16
local
active



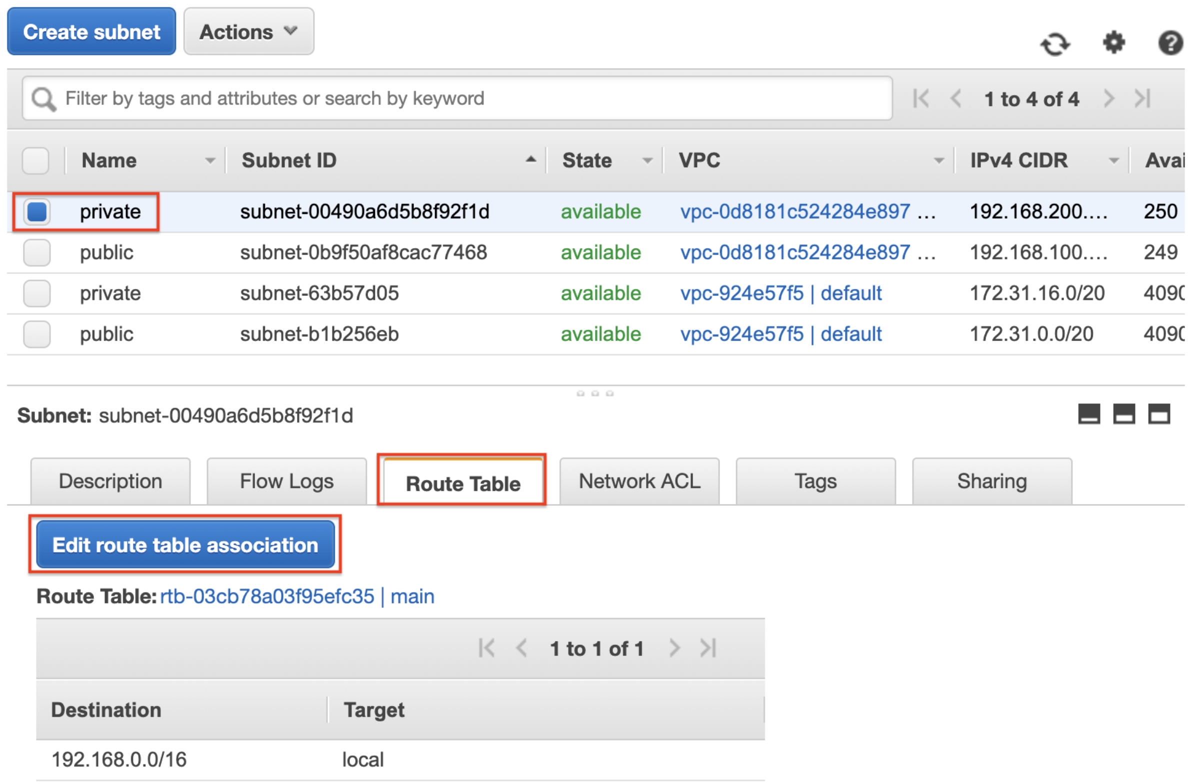


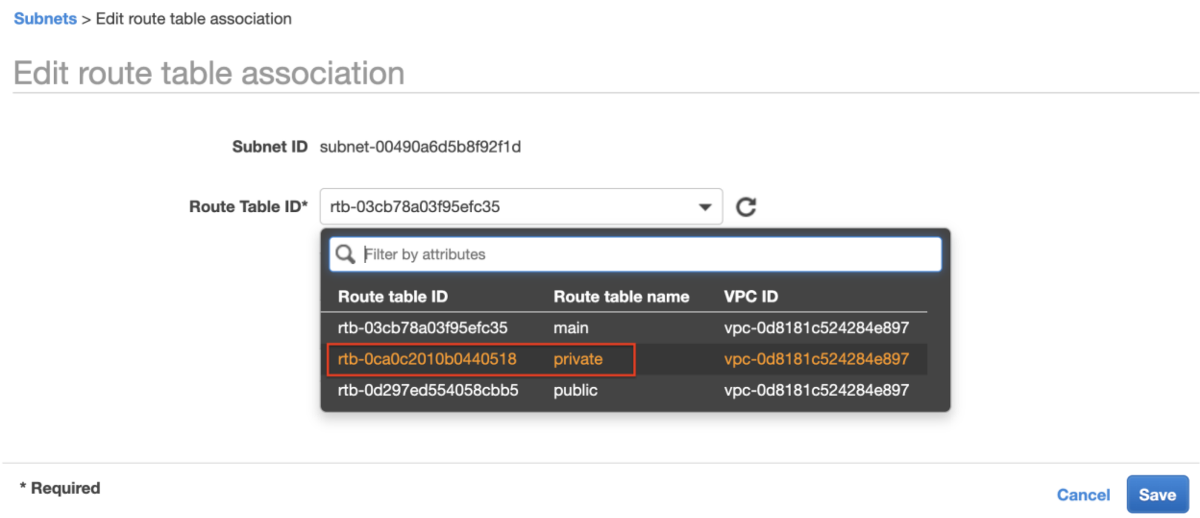


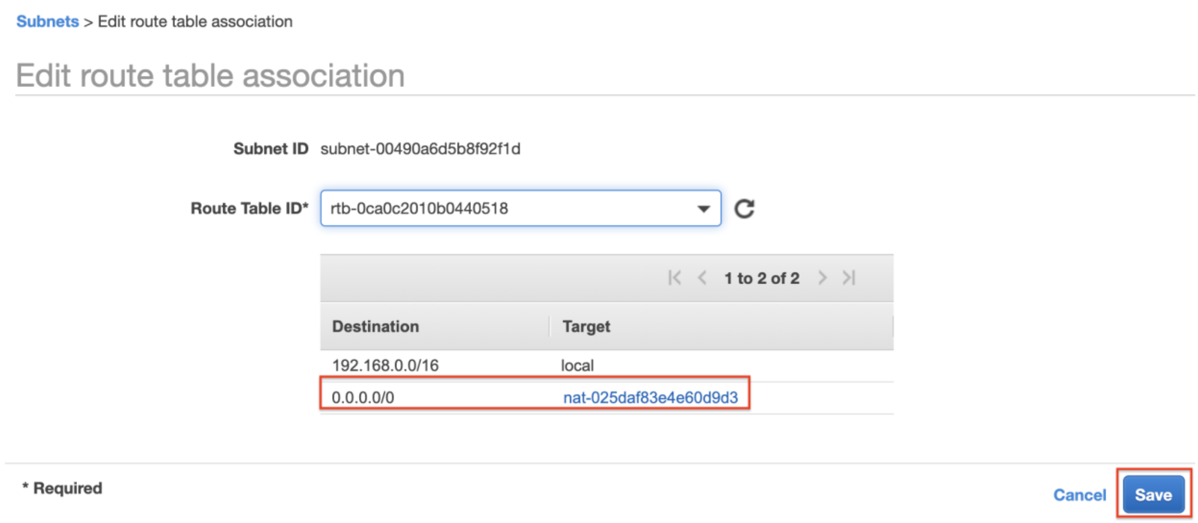


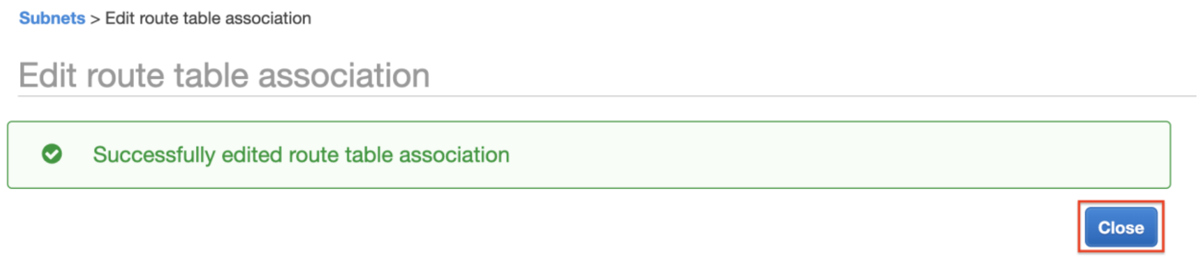
Step 4 - Associate the Route Table to the Private Subnet

Associate the newly created route table to the private subnet.


Create subnet
Actions
Filter by tags and attributes or search by keyword
1 to 4 of 4
Name
Subnet ID
State
VPC
IPv4 CIDR
Avai
private
subnet-00490a6d5b8f92f1d
available
vpc-0d8181c524284e897 ...
192.168.200....
250
public
subnet-0b9f50af8cac77468
available
vpc-0d8181c524284e897 ...
192.168.100....
249
private
subnet-63b57d05
available
vpc-924e57f5 | default
172.31.16.0/20
public
subnet-b1b256eb
available
vpc-924e57f5 | default
172.31.0.0/20
Subnet: subnet-00490a6d5b8f92f1d
Description
Flow Logs
Route Table
Network ACL
Tags
Sharing
Edit route table association
Route Table: rtb-03cb78a03f95efc35 | main
1 to 1 of 1
Destination
Target
192.168.0.0/16
local


Subnets > Edit route table association
Edit route table association
Subnet ID subnet-00490a6d5b8f92f1d
Route Table ID* rtb-03cb78a03f95efc35
Filter by attributes
Route table ID
Route table name
VPC ID
rtb-03cb78a03f95efc35
main
vpc-0d8181c524284e897
rtb-0ca0c2010b0440518
private
vpc-0d8181c524284e897
rtb-0d297ed554058cbb5
public
vpc-0d8181c524284e897
* Required
Cancel
Save





Test and Validation

To validate this lab you need to access the EC2 instance in the private subnet (labeled as B) and from there try to access the internet, for example:

```
curl www.google.com
```